



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Database Mgmt & Warehousing				
Course Code:	DBA1000	Credit Value:	4	Credit Hours:	56
Programs:	BAGC I.T. Business Analysis BAPG Business Analytics CAGC Crime Analytics HAGC Health Analytics MAPD Mobile Application Development				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Degree Breadth Course: ☐

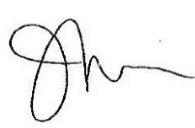
Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will explore fundamental concepts in data and information management. The course content centers on the essential skills of identifying organizational requirements, modelling requirements using conceptual data modelling techniques, converting conceptual models into relational data models, and verifying structural characteristics with normalization techniques. Students will focus on the Structured Query Language (SQL) to define and manipulate data. Additionally, students will explore the data warehouse lifecycle and dimensional modelling for data warehousing.

Date: June 21, 2021

Prepared By: Nathan Abourbih

Approved by: 

JILL FERGUSON
Chair, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation.
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Recommend different systems architectures and data storage technologies to support data analytics. 3. Design and apply data models that meet the needs of a specific operational/ business process. 4. Develop software applications to manipulate data sets, correlate information and produce reports. 5. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 6. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions. 7. Implement data solutions in compliance with corporate policies, ethical standards, and industry regulations.
Crime Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Develop a crime analytics methodology grounded in a conceptual framework and consistent with criminal intelligence techniques.
Health Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Design health indicators and apply data models that meet organizational, program and service needs. 3. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 4. Apply health analytics and business intelligence tools to support evidence-informed practice. 5. Assess organizational requirements for health information needs and evaluate the impact on the organization, programs and services, and on health services delivery.

Program(s)	Vocational Learning Outcomes
	6. Develop, implement, and evaluate health information management practices, policies and processes to support client and community care, organizational goals, operations, and regulatory compliance. 7. Ensure compliance with the legal health obligations, as well as with the professional, ethical and organizational standards that ensure privacy, security and confidentiality in the access, retention, storage and disposal of personal health information. 8. Ensure the completeness, accuracy, consistency, timeliness and integrity of health information throughout the management of its lifecycle. 9. Work professionally, ethically and collaboratively with stakeholders and as a member of an interdisciplinary health team, to enhance the collection, distribution, use, security and awareness of quality health information and its impact on health.
Mobile Application Development	College Standards 1. Evaluate business and design requirements to select, formulate and implement mobile solutions. 2. Select and integrate database and server-side technologies into mobile solutions. 3. Design, develop and build a database to application specifications.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Relational Databases
- Structured Query Language (SQL)
- Database Normalization
- Database Design Process
- Data Modeling
- Data Warehousing and Dimensional Modeling

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Discuss the fundamental concepts of data and information management.	1.1 Investigate examples of databases used in industry today. 1.2 List and define the components of a database. 1.3 Define what a database management system (DBMS) is and list its properties. 1.4 Discuss the characteristics and benefits of a database. 1.5 Discuss the evolution of data storage and processing from the use of simple files, to relational databases, to big data solutions.

Learning Outcome	Objectives
2. Use the SQL Data Manipulation Language (DML) to query databases.	2.1 Describe the purpose of querying a database. 2.2 Write SELECT statements to retrieve data from a single table and multiple tables. 2.3 Create queries that use comparison and logical operators. 2.4 Create queries that use scalar and aggregate functions. 2.5 Create queries that group and summarize data. 2.6 Investigate examples of complex query development.
3. Use the SQL Data Definition Language (DDL) to create databases and work with objects within databases.	3.1 Create queries that insert, update, and delete data. 3.2 Write queries that create, modify, and delete databases, tables, constraints, as well as indexes.
4. Identify and describe the relational database model.	4.1 List and describe basic terminology relating to the relational database model. 4.2 Describe and identify functional dependencies, determinants and dependent attributes. 4.3 Recognize and define the use of primary, candidate, and composite keys. 4.4 Identify anomalies in a relationship.
5. Explain database normalization and how it is used in database design.	5.1 Describe what normalization is and why it is important in the database design process. 5.2 List and describe normalization categories, forms, and various design principles. 5.3 Normalize an unnormalized data structure to 3NF. 5.4 Identify when denormalize is appropriate and describe the process of denormalization.
6. Analyze organizational requirements to create a conceptual database model.	6.1 Describe the purpose of the data modelling process. 6.2 Describe two main types of conceptual database models: ERD and UML. 6.3 Illustrate the use of various conceptual modelling techniques to identify data entities or domain classes. 6.4 Identify entities, relationships, and cardinality. 6.5 Discuss and describe various complex relationship types associated with UML diagramming: generalization/specialization and whole-part relationships.
7. Transform conceptual data models into logical and physical database designs.	7.1 Describe how to transform a data model into a database design. 7.2 Identify the differences between a conceptual, logical, and physical database model. 7.3 Create a logical database and physical model. 7.4 Identify required attributes and data types. 7.5 Identify primary keys and describe when to use a surrogate key.

Learning Outcome	Objectives
	7.6 Describe when and how to use referential integrity constraints. 7.7 Create a functioning database from a physical database design.
8. Describe the basic concepts of data warehousing and the data warehousing lifecycle.	8.1 Describe the purpose of data warehouses and data marts. 8.2 Describe the basic concepts of data warehousing. 8.3 List and describe the components of a data warehouse. 8.4 Describe how business intelligence applications interact with data warehouses and operational databases. 8.5 Discuss the data warehousing lifecycle.
9. Define the fundamental concepts of dimensional modelling for data warehousing.	9.1 Describe the purpose of a dimensional database. 9.2 Describe the dimension design process. 9.3 Investigate and describe various examples of the star and snowflake schemas. 9.4 Describe the basic concepts of online analytical processing (OLAP) and data mining.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Presentations
- Guest Speaker(s)
- Reflective Writing
- In-Class Exercises
- Case Studies

Resources Required

Additional Supplies

Minimum Technology Requirements (BAGC/MAPD)

A personal laptop computer that meets this program's minimum technology requirements and regular access to a reliable Internet connection.

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities (G)	40%
Quizzes	20%

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>